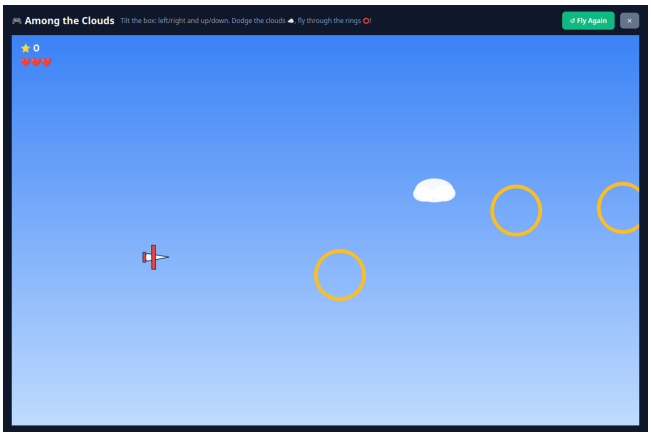
Among the Clouds

A game about tilt, despite the cloudy title — and an aeroplane that flies inverted when the box does

Level: Junior and Explorer — ages 7–15 (Romanian school years II–VIII) • **Time in class:** 15–20 min, as a hook or a station • **Sensors:** MPU-6500 (acceleration)



The activity screen. The values in the picture are simulated; your own will differ.

Before you start

Who it is for

Junior and Explorer — ages 7–15 (Romanian school years II–VIII). As a game it works at any age. **As a lesson the subject is not flight but tilt**: with younger children it is an intuitive way into the idea of a position sensor; from year VI you can discuss *why* the box knows how it is being held. The curricular proposal classifies it as an *optional hook*, not a core activity — use it as an opener or as a 15–20 minute station.

Assumed knowledge

Nothing. It is playable from the first second. For the discussion afterwards: the idea that gravity always pulls downwards.

Time in class

15–20 min, including the debrief. **Do not stretch it**: it works as a hook but has not enough content for a full lesson — and the curricular proposal warns that using all three games in the zone eats the time meant for the central investigation.

Teacher preparation

5 min — play it once yourself, to see how quickly three lives go; and **check a box lying flat**: if the aeroplane appears inverted or flies tail-first, that box either has old firmware or has its sensor mounted the other way up — see the technical note.

Sensor settling time

None. The accelerometer responds instantly.

Sensors and boxes

MPU-6500. **The sensor is mounted upside down in the box and the firmware already corrects for it**: a flat box reports $AZ \approx +1$ g. **In this activity the sign genuinely matters** — see the technical note. One box is enough, with the screen projected and turns taken.

Three routes

Short	15–20 min: two or three rounds of 60–90 seconds, taking turns, then the debrief — which is the part that justifies the lesson.
Full	As a station in a carousel lesson: 15 minutes alongside two other stations, with the same three debrief questions at each group.
Project	Extension: the "what controls what" test — hold the box still and tilt in one direction only, to separate roll from pitch.

1. Where this fits in the curriculum

The title misleads, and it is worth saying so at the start: the activity is not about clouds, weather or aviation. **It is about a tilt sensor**, and the clouds are scenery.

- **Mathematics and exploring the environment** (years I–II) — orientation in space, left and right, up and down, and the link between a movement of the hand and a visible effect.
- **Natural sciences** (years III–IV) — gravity as a force that always pulls downwards; the game makes it tangible, because the aeroplane "knows" which way is down.
- **Physics** (from **year VI**) — **this is the entry that justifies the activity at lower-secondary level**: how can a closed instrument with no windows know how it is being held? The answer — through the direction of gravity — is short, exact and surprising, and it is put into words; computing the angles with trigonometric functions belongs to year VII and above.
- **Physical education and motor skills** — fine control and hand-eye coordination; the game asks for small continuous movements, not sudden ones.
- **Cross-curricular** — the idea of an *interface*: a physical object becomes a control. It is a first encounter with the idea that a sensor can replace a button.

The core skill, at the debrief, is **working out what controls what**. A child who can say "tilting sideways moves the aeroplane sideways, and tilting forwards and back makes it climb and descend" has extracted a rule from a system by trial — which is the method itself.

2. Time plan

Route	Time	What happens
Hook	**15–20 min**	The prediction (2), a demonstration round with the screen projected (3), two or three rounds in turn of 60–90 seconds (8), the "what controls what" test (3), the debrief (4).
As a station	15 min	In a three-station lesson: play, fill in the "what controls what" table, answer the three debrief questions.

Short rounds, and everybody with a job

With one box and thirty children, the game can become twenty-nine children watching. **Rounds are capped at 60–90 seconds** — three lives go quickly anyway — and **every waiting child has a named role**:
the controls observer watches what the player does with the box and what the aeroplane does, and fills in the "what controls what" table; **the timekeeper** calls the end of the round; **the scribe** records the score and, more importantly, *what the player changed* since the previous round.
The roles rotate every round, so every child both plays and observes. A child who has observed two rounds before playing does visibly better — and that is itself an observation to make at the debrief.

3. Materials

Item	Per group	Notes
MagicBox with MPU-6500	1	It becomes the control stick.
Laptop with the app	1	**WiFi is far better here**: on a cable, moving the box pulls the cable and the cable pulls back.
Lanyard or wrist strap	1	**Recommended.** The box moves a great deal and quickly during play; a strap prevents a drop.
Clear space around the player	—	A metre in every direction. Players turn without meaning to.
Projector	1	So that the others can see and observe.
Stopwatch	1	For rounds of 60–90 seconds.

Preparation. Play a round yourself. And **check a box lying flat**: the aeroplane should appear upright, nose to the right. **If it appears inverted or flies tail-first**, that box has either old firmware or its sensor mounted the other way up — see the technical note: the two have different remedies.

4. Scientific background for the teacher

How the box knows the way it is being held

The accelerometer has no compass and no windows. It measures **acceleration on three axes**, and at rest the only acceleration present is gravity: a vector pointing always downwards.
When the box rotates, gravity stays where it is, but *the box's axes* rotate relative to it. From how much of the vector falls on each axis, the tilt is recovered:

$$\text{roll} = \arctan\frac{a_y}{\sqrt{a_x^2 + a_z^2}}, \qquad \text{pitch} = \arctan\frac{-a_x}{\sqrt{a_y^2 + a_z^2}}.$$

These are the game's two controls: **roll moves the aeroplane sideways, pitch makes it climb and descend**, with nose up meaning climb. The formulas are for the teacher; the children read the two angles straight off the screen.
The limitation, and it is felt in play: the method works only while the box is not *accelerated* by anything other than gravity. A sudden movement of the arm adds its own acceleration, and the aeroplane lurches for a moment. That is why the game is played with small continuous movements — and why a real aircraft uses gyroscopes as well as accelerometers.

4.1 Why the aeroplane flips — and why the sign matters

The rotation of the aeroplane *sprite* does not use the same formula as the control. It uses

$$\text{roll}_{\text{full}} = \arctan_2(a_y,\, a_z),$$

the two-argument arctangent, which covers the whole circle from $-180°$ to $+180°$. So if you turn the box upside down, **the aeroplane flies inverted** — which is a good joke and, besides, the most memorable part of the activity.

But this is also where the one possible fault comes from. Unlike roll and pitch, where a_z enters squared and its sign disappears, **here a_z enters directly**. Its sign matters.

The sensor is mounted upside down in the box, and **the firmware already corrects for it**: a flat box reports $AZ \approx +1$ g and the sprite appears upright. A box that shows -1 g flat makes the aeroplane appear inverted or fly tail-first — and, for the same reason, climb when the nose goes down. **It has two possible causes, with different remedies:**

- **Firmware older than v2.2.1.** The app flags it when the box connects (**Old firmware**). The remedy is to reflash the box.
- **Current firmware, but the sensor fitted the other way up** — not every box is built the same. The remedy is the **Motion sensor upside down** tick under **Settings → Box names**, set once for that box; the app then turns its motion data the right way up in every activity and in its files. With firmware 2.3.0 or newer the tick is kept in the box itself, so it counts on every computer; the box must be connected when you tick it.

Do not "fix" this anywhere else. The Settings tick concerns one box; a correction made for one game would break every box that works correctly — which is exactly what happened once in this project's history.

4.2 What kind of game it is, as a system

Worth knowing for the debrief. The obstacles drift leftwards, ever faster — the speed goes up a little every five rings flown through —, and the aeroplane moves sideways and vertically in proportion to the tilt — **so tilt controls velocity, not position**. This is a real and counter-intuitive distinction: to stay put, you must hold the box level, not hold it in a particular place.

Children discover it empirically, and putting it into words at the debrief — *how far I tilt decides how fast I move, not where I end up* — is the most mature idea the activity can produce. It is, incidentally, the difference between steering a car and dragging an object with a mouse.

5. Lesson flow

1. **The prediction (2 min).** Before playing: *how do you think the aeroplane is controlled — by pressing something, by moving it through the air, or by tilting it? And what do you think happens if you turn the box upside down?* The second question is the good one, and almost nobody predicts the answer.
2. **A demonstration round, projected (3 min).** Play a round yourself, with the class watching, without explaining the controls. **Let them work out** what each movement does — observing is the task.

Technical note — the controls, the rules and the sign of AZ

The sensor. MPU-6500. There are no cards and no chart: **the whole screen is the game**. In the top-left corner, on a dark plate, sit: the score (the star) and **Best**, the best score; the lives (three hearts); **the two angles shown live**, for example *Roll* $-12°$ · *Pitch* $5°$ — with them the "what controls what" test can be recorded in degrees too —; and the **Speed** line, which starts at $\times 1.00$. From firmware 2.2.5 the chip smooths its own readings at about 20 Hz, so the aeroplane trembles less.
How it starts. From the **Activities** page, from the **Among the Clouds** card, with the **Start** button. A box is chosen when it opens.
The controls. Roll — tilting sideways — moves the aeroplane sideways; **pitch** — tilting forwards and back — makes it climb and descend, *nose up means climb*. **Tilt controls speed, not position**: at full tilt, about $30°$, the aeroplane moves at 260 pixels a second, and with the box level it stays put.
The rules. Three lives. Obstacles come from the right, and **every five rings flown**

through the game speeds up: the speed rises by 15 % of the starting speed — the **Speed** line reads ×1.15, then ×1.30 after ten rings —, obstacles come more often, and **Faster!** flashes briefly at the top. The rise stops at ×2.20, after forty rings. About **a third are rings** — fly through them for a point — and **two thirds are clouds**, to be dodged. After a collision you have **one and a half seconds of invulnerability**, which explains why you sometimes pass through a cloud without losing anything. **The score is the number of rings**, not the time survived, and **the best score stays on screen** from one round to the next, and even from one day to the next on the same computer. At the end **End of flight!** appears, with the number of rings and the best score (or **New best score!**), and the **Fly Again** button at the top starts a new round.

The sign of AZ matters here — the only activity where it does. The sprite's rotation uses $\arctan_2(a_y, a_z)$, so it depends directly on the sign of a_z. **A flat box must give** $AZ \approx +1$ **g**, and the aeroplane must appear upright, nose to the right. **If the aeroplane appears inverted or flies tail-first**, first look whether the app flagged **Old firmware** when the box connected: then the box has firmware older than v2.2.1 and is reflashed. If the firmware is current, the box has its sensor fitted the other way up: tick **Motion sensor upside down** under **Settings → Box names**. The game itself has no correction of its own, and must not have one.

WiFi, not cable. The box rotates a great deal; on a cable it twists and pulls. Over WiFi the game is noticeably better.

3. **The rounds (8 min).** In turn, 60–90 seconds each, with the roles from the tip box for those waiting. **Do not announce the scores as a league table.**
4. **The "what controls what" test (3 min).** Hold the box still and tilt it *only* one way, then *only* the other. The children fill in the table: what I tilted, what the aeroplane did. **This is the investigation part of the activity.**
5. **The debrief (4 min).** See the box below. It is the part that turns the game into a lesson, and it is not skipped.

Debrief: the score is not the point

Do not ask "who got the highest score". The score depends largely on the luck of where the rings appear, and a league table turns an observation into a competition. The best score the app keeps is there so that each child can compare with themselves, not with the neighbour.

The three questions, in order:

- *What controls what?* — the full answer is: tilting sideways moves the aeroplane left and right, tilting forwards and back makes it climb and descend, and nose up means climb. Ask for it in their own words, checked against the isolation test.
- *Does how far you tilt decide* ***where*** *you end up or* ***how fast*** *you move?* — this is the hard question and the most valuable one. The answer is speed. A child who reaches it unaided has understood an idea about control systems.
- *How does the box know how you are holding it?* — gravity. It has no windows and no compass; it feels which way the weight pulls.

And a fourth, if it has come up: *why does the aeroplane fly inverted when you turn the box over?* Because the sprite follows the full rotation, not just the small tilt — and that can be shown by rotating the box slowly with the screen projected.

The score is not the objective. A child with a low score who can explain the three answers above has learned everything the activity had to teach. Say so *before* the first round, not after — otherwise the game becomes a contest and the debrief a formality.

The investigation, in one table

The question. Which movement of the box controls which movement of the aeroplane?
The hypothesis. Written **before** playing: how do you think it is controlled — by pressing, by moving through the air, or by tilting?
What I change. One tilt at a time: first sideways only, keeping the box level fore and aft; then fore and aft only, keeping it level sideways.
What I observe. The direction the aeroplane moves, **whether it stops** when I bring the box back to level, and the angle shown on screen for each tilt.
What I keep the same. The box *does not travel through the air* — it is held in one place and only tilted. That is the condition of the whole test: if you move it, you add acceleration and mix the causes.
How many repeats. Each direction twice, to be sure it was not a coincidence of the game.
What makes an observation valid — and what makes it an artefact. Valid is an observation made with the box held in place and tilted smoothly. An artefact is anything else: **a sudden movement of the arm makes the aeroplane lurch**, because the accelerometer cannot tell the acceleration of the movement from gravity; and a twisted cable pulls the box and adds a tilt nobody made.

6. Analysis after the activity

6.1 What the observations should look like

What I do with the box	What the aeroplane does	What it means
Hold it level	Stays put vertically	Tilt controls speed; zero tilt, zero movement.
Tilt it to the left	Goes left	Roll.
Tilt it towards me (nose up)	Climbs	Positive pitch.
Tilt it further	Moves *faster*, not further	**The game's central idea.**
Turn it upside down	Flies inverted	The sprite follows the full rotation.
Move it sharply through the air	Lurches uncontrollably	The accelerometer confuses movement with gravity.
Flat box, current firmware	Aeroplane upright, nose right	This is how it should look.
Flat box, old firmware	Inverted / tail-first	**Reflash the box.**
Flat box, current firmware, still inverted	Inverted / tail-first	Sensor fitted the other way up: **the Settings tick** for that box.

6.2 Questions to ask afterwards

1. The box has no windows and no compass. How does it know how you are holding it?
2. When you tilted further, did the aeroplane go further or move faster? Which of the two is true, and how did you check?
3. Why does the game go wrong if you move the box sharply through the air, even though the tilt angle has not changed?
4. You once flew through a cloud without losing a life. Why do you think that happened?

5. If you were designing the game, would you make tilt control speed or position? Which would be easier to play, and why?

6.3 Assessment ideas

- **Low stakes:** the child completes the "what controls what" table with arrows rather than words.
- **Standard:** the child describes both controls in full sentences and explains how the box knows the way it is held.
- **Extension:** the child argues, from their own observations, that tilt controls speed rather than position — and proposes a test that would tell the two apart.

Common pitfalls and misconceptions

Turning it into a league table. The score depends heavily on the luck of the rings. A ranking moves attention from observation to competition.
Walking the box through the air. Children instinctively try to "carry" the aeroplane by moving the box. It does not work and it spoils the observation: the box is held in place and only tilted.
A twisted cable. The box rotates a lot; on USB the cable tangles and pulls. Work over WiFi.
"The aeroplane is broken" — when in fact it is flying inverted. If it flies upside down with the box flat, it is either old firmware (reflash the box) or a box with its sensor fitted the other way up (tick **Motion sensor upside down** in Settings). Nothing else changes.
Rounds that run too long. With one box, a five-minute round means twenty-nine bored children. 60–90 seconds, with roles for the rest.
Skipping the debrief. Without it the activity is only a game — and the curricular proposal classifies it as an optional hook precisely because it risks consuming time without leaving anything behind.
Using all three games in the zone. Choose *one*. All three together eat the time meant for the seismic investigation, which is the zone's core.

Safety and classroom management

A lanyard or wrist strap is recommended. The box moves a great deal and quickly; a sweaty hand or an enthusiastic gesture can throw it. If you have no strap, the player sits down with the box held in both hands over the table. The box is never thrown or shaken near classmates — not even to "help" the aeroplane.
A metre of clear space around the player. Children turn without meaning to and hit classmates or furniture. Agree the area before the first round.
The player stays put. No walking, no running, and no playing standing on a chair. All the control is done from the wrist.
Cables. If you do play over USB, the cable can tangle around the arm. WiFi is both safer and better for the game.
Glasses and objects on the table are moved aside before a round — a wide gesture sweeps the desk.

Adaptations for pupils with special educational needs

The observer's role is equal to the player's. This is the main adaptation and it must be said to the class: **the "what controls what" table is completed from observation**, and whoever observes two rounds usually answers the debrief better than whoever plays. A

child who cannot or will not play loses nothing of the lesson.
The box resting on the table. For children with difficult fine control, the game can be played with the box *resting on the edge of the table* and tilted there, rather than held in the air. The control is coarser, but the game works.
Three real roles, rotated every round. The controls observer fills in the table; **the timekeeper** calls the end of the round; **the scribe** records what the player changed since the previous round.
The movement on screen. A continuously scrolling background can be unpleasant for children sensitive to motion or with vestibular difficulties. **Seat them further from the screen or to one side**, give them the scribe's role — which does not require a fixed gaze — and cap the rounds at 60 seconds.
Frustration. Three lives go quickly at first. Say in advance that the first rounds are for *finding out the controls*, not for the score.

If things do not go to plan

The aeroplane flies inverted or tail-first. If the app flagged **Old firmware**, the box has firmware older than v2.2.1: **reflash it** — see the flashing guide in the package. If the firmware is current, the box has its sensor fitted the other way up: under **Settings → Box names** tick **Motion sensor upside down** for it. The tick concerns that box only; correct boxes stay correct.
The aeroplane lurches uncontrollably. Somebody is moving the box through the air instead of tilting it. That is the activity's own observation — show it to the class and ask for the explanation.
The game stutters. On USB with a twisted cable, or on weak WiFi. Check the signal; the game needs frequent data.
One box and thirty children. That is the usual case. Rounds of 60 seconds, screen projected, roles for everybody — see the tip box.
You have no time for the whole activity. Keep the prediction, one round and **the three debrief questions**. Without them the fifteen minutes leave nothing behind.
The box will not connect. Do not stop the lesson — but do not replace it with something else either: this is an optional opener, and if it will not run, move to the zone's central investigation (*The Seismometer*), which has content of its own. Play another day, after checking the box and the network.

7. Further resources

Stable, well-established starting points. Search within each site for the topic named — these are base addresses, not deep links, so they do not rot.

- **Institute of Physics** — `https://www.iop.org` — search for material on forces and gravity, for the answer to "how does the box know".
- **PhET Interactive Simulations** — `https://phet.colorado.edu` — search for the forces and motion simulations; useful for the difference between controlling speed and controlling position.
- **STEM Learning** — `https://www.stem.org.uk` — search for sensors and control systems for lower secondary.
- **NASA** — `https://www.nasa.gov` — search their pupil material on flight controls, if the activity provokes questions about real aircraft.
- **Royal Aeronautical Society** — `https://www.aerosociety.com` — the principles of flight, for the natural continuation into *Flying Straight*.

8. Assessment

The exit ticket, identical to the one on the pupil sheet: *how does the box know the way you are holding it?* Collected in the last two minutes and, in this activity, far more informative than the score.

For a more formal judgement, use the **common three-axis rubric** — *measurement practice*, *data interpretation* and *scientific reasoning*. Here the three axes look like this:

- **Measurement practice** — in the isolation test, the child holds the box in place and tilts one direction at a time.
- **Data interpretation** — the child fills in the "what controls what" table correctly and notices that the aeroplane stops when the box is level.
- **Scientific reasoning** — the child explains how the box knows the way it is held, and argues that tilt controls speed rather than position.

The rubric itself, with descriptors for all three levels, is defined once in the *curriculum plan* and is not repeated in the guides.

9. Pupil sheet

The sheet the children fill in is a separate document: `docs/en/student_sheets/among_the_clouds.pdf`. It has two pages — the prediction and the "what controls what" table on the first; the rounds table, the isolation test and the conclusion on the second — so it is printed double-sided, one sheet per child.

Three things to know before printing it. **The "what controls what" table is completed with arrows rather than words**, so that it can be done by younger children and while somebody else is playing. **The rounds column asks for "what I changed", not only the score** — without it the game stays a game. **The sheet has no room for a class league table**, and that is a decision: the score depends heavily on the luck of the rings, and a column for "my place" would move attention exactly where it should not go. The tables are not repeated here precisely so that the two documents cannot drift apart: there is one printable version, and it is the sheet.